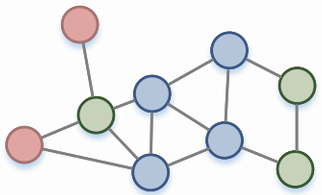
Input: Graph and Node Features



$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \boldsymbol{X})$

1. Class-space Anchoring

Node features $\boldsymbol{X}$ $N \times d$

Linear projection $\boldsymbol{W}$ $d \times C$

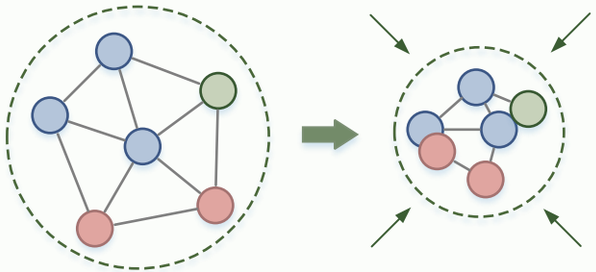
Initial class-state matrix $\boldsymbol{Z}(0) = \boldsymbol{X}\boldsymbol{W}$ $N \times C$

Node state in class space



2. Continuous Velocity Field in Class State Space

(a) Attractive Velocity (Smoothing / Contraction)



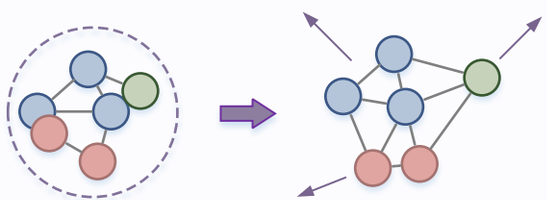
$\boldsymbol{S}_+(\boldsymbol{Z}(t)) = -\boldsymbol{L}\boldsymbol{Z}(t)\boldsymbol{K}$

$\boldsymbol{K} = \kappa \boldsymbol{I}_C + Diag(softplus(\alpha))$

$\boldsymbol{K} \succcurlyeq \kappa \boldsymbol{I}_C,\ \kappa > 0$

Satisfies Assumption 1

(b) Local Separation Correction (Repulsion / Divergence)



$$\boldsymbol{s}_{i,-}(t) = \sum_{j \in \mathcal{N}_i} a_{i,j} \frac{\boldsymbol{z}_j(t) - \boldsymbol{z}_i(t)}{\|\boldsymbol{z}_j(t) - \boldsymbol{z}_i(t)\|^2 + \epsilon},$$

$\epsilon > 0.$

Satisfies Assumption 2

Satisfies Assumption 3

3. Stability-constrained Velocity Regulation

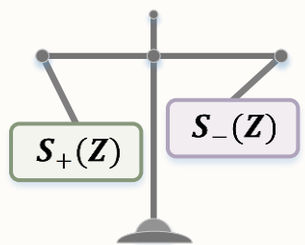
Correction strength

$$\mu = \frac{(1-m)\kappa\lambda_2(\boldsymbol{L})}{\hat{\rho}_-},\quad m \in (0,1)$$

$\mu\hat{\rho}_- < \kappa\lambda_2(\boldsymbol{L})$

Satisfies Assumption 4

$\boldsymbol{S}_-(\boldsymbol{Z}) = [\boldsymbol{s}_{1,-}{}^{\mathrm{T}}, \boldsymbol{s}_{2,-}{}^{\mathrm{T}}, \ldots, \boldsymbol{s}_{N,-}^{\mathrm{T}}]^{\mathrm{T}}$



$\dot{\boldsymbol{Z}}(t) = \boldsymbol{S}_+(\boldsymbol{Z}(t)) - \mu\boldsymbol{S}_-(\boldsymbol{Z}(t))$

4. ODE Solver

$\boldsymbol{Z}(0)$

$$Z(T) = Z(0) + \int_0^T [-\boldsymbol{L}\boldsymbol{Z}(\tau)\boldsymbol{K} - \mu\boldsymbol{S}_-(\boldsymbol{Z}(\tau))]\, d\tau$$

Solve ODE in the class state space

$\boldsymbol{Z}(T)$

5. Prediction

$\hat{Y} = softmax(\boldsymbol{Z}(T))$



Node classification